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Application of process mining to support simulation model development: A use case in production planning

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Abstract

The increasing complexity of industrial production systems requires efficient methods for model development in simulation-based planning processes. Conventional approaches to building discrete-event simulation models are often associated with significant effort for data preparation and modeling, which in practice frequently represents a barrier to the effective use of simulation. This paper demonstrates how Process Mining can be used to address these challenges and to support the development of data-driven simulation models in production and logistics. For this purpose, a generic data-driven procedure model is outlined, describing the structured transformation of event logs into simulation-ready models. In an initial prototypical use case from a production context, it is examined to what extent and with what quality data can be extracted from event logs using Process Mining to make them usable for simulation-based planning. Finally, the developed approach is methodically evaluated regarding its applicability and added value in the context of simulation-supported production planning. The results illustrate that Process Mining can reduce modeling effort and improve data quality, provided that an appropriate data foundation is available. Furthermore, the approach opens new potentials for objective and iteratively adaptable process design in dynamic production environments.

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1. Introduction

The simulation of production and material flow systems has proven to be an established instrument in production planning, as it enables cost-efficient, low-risk and at the same time insightful analyses of complex value creation systems [1]. In particular, discrete-event simulation supports planning and control by rendering the effects of individual system components transparent, thereby enhancing the understanding of interdependencies in volatile and dynamic environments [2,3]. The academic discourse likewise emphasizes simulation as a valuable tool for the analysis and planning of complex production systems. At the same time, numerous studies highlight that major challenges reside in reducing the time required to conduct simulation studies and in achieving a stronger integration of simulation into real production environments [4,5,6]. With the growing availability of

extensive operational data generated by IT systems and sensors in modern production facilities [7], there is also an increasing demand to harness these data efficiently for simulation-based analyses.

At this intersection, process mining comes into play: as a data-driven analytical approach, it extracts real process executions from event logs, reconstructs process structures, variants, and performance indicators [8,9], and thereby provides a robust foundation for data-driven simulation models. At the same time, process mining makes deviations and bottlenecks in the as-is process transparent, supports model development, and systematically uncovers optimization potentials along operational workflows, thus bridging the gap between model-driven planning and data-based process reality [9,10].

According to this rationale, the integration of both approaches can reduce the effort required for reconstructing and

modeling material flows, calibrate parameters against observed behavior and iteratively adapt simulation models to real process changes. Against this backdrop, the contribution presents an industrial case study demonstrating how event data from a production process can be processed through process mining to generate simulation-ready representations for production and logistics. Initially, no knowledge was available regarding the specific use case, the production process, the material flow, or the machines involved. Accordingly, the preparation of event data was carried out in a purely data-driven manner; the process models and parameters derived were subsequently compared with and validated against the real process. At the same time, the analysis identifies where and why common process mining tools encounter data- or method-related limitations. This contribution represents the starting point of a series of case studies on the proposed approach, which will be gradually expanded. In doing so, it advances the field of data-driven design of production systems by systematically employing event data for model development and parameterization.

2. Background

2.1. Process Mining in Manufacturing

The scientific discourse on process mining has thus far been situated predominantly within the fields of business process management and computer science [10]. Originally developed for the analysis of traditional business processes, the approach is increasingly applied in industrial contexts such as manufacturing, where it enhances transparency and reveals optimization potential [11]. Empirical studies further demonstrate that process mining leads to measurable improvements in operational performance and thereby contributes to efficiency gains while also enabling significant cost reductions [12]. More recently, scholarly attention has shifted towards the production domain: here, it becomes evident that relevant data sources are not limited to ERP and MES systems, but increasingly include OT/IoT sensor and machine data, thereby placing the interoperability between IT and shop-floor systems at the center of attention [13].

An empirical case study by Lorenz et al. [14] in discrete make-to-stock manufacturing demonstrates that, through the consistent use of event logs from manufacturing information systems, process mining can be employed to uncover production-oriented optimization potentials. Exemplary contributions in this field range from Aslan et al. [15], who utilize positional data of employees as event logs, to Lugaresi and Matta [16,17], who investigate (semi-)automated approaches to digital twin and simulation model generation.

The review by Oliveira et al. [18] consolidates such examples to systematically elaborate both the potentials such as increased transparency, bottleneck and variability analysis, online prediction, and coupling with digital twins and simulation and the pitfalls, including heterogeneous data spaces, missing or inconsistent logs, case and cycle detection, and governance challenges. At the same time, typical data-related issues are described as practical barriers, among them fault-tolerant case construction, time synchronization across machines, heterogeneous log granularities along the ISA-95

levels, and the fragmentation of information across multiple systems. As a key methodological advancement, object-centric process mining (OCPM) is highlighted, which transcends the single-case perspective by linking events to multiple objects (e.g., order, lot, material, resource) [19].

2.2. Process Mining and Simulation

Research on the integration of process mining and simulation has steadily expanded in recent years, covering an increasingly broad spectrum of applications. These range from the incorporation of predictive maintenance strategies in business process management [20], through hybrid, domain-specific approaches in healthcare [21], to initial applications in production planning [22]. Fundamentally, the coupling of process mining and simulation serves two complementary functions: on the one hand, it enables the log-based derivation and parameterization of robust simulation models; on the other, it supports forward-looking what-if analyses for the evaluation of redesign options [23].

Van der Aalst [24] explicitly positions simulation as a forward-looking complement to process mining, opening methodological opportunities to address what-if questions and thereby extending beyond purely descriptive or diagnostic analyses. In line with this perspective, Nakatumba et al. [25] were among the first to demonstrate that linking process mining with discrete-event simulation can enhance the understanding, modeling, and improvement of real processes. Building on this foundation, Castiglione [26] pursues an event-centered approach in which structural and parametric elements of simulation-ready factory models such as Petri nets, routing probabilities, arrival and processing time distributions, and resource allocations are derived from and calibrated against log data. Bemthuis and Lazarova-Molnar [27] place a complementary emphasis on the systematic integration of process mining and agent-based modeling and simulation (ABMS). They develop a conceptual framework that specifies the role of process mining across the entire ABMS lifecycle from behavior-based reconstruction and initialization, through parameterization, conformance checking, and calibration, to continuous validation, while simultaneously identifying domain-specific research gaps.

Regarding production and logistics environments, Özkul et al. [28] introduce the concept of simulation-enhanced, action-oriented process mining. In this approach, streaming event logs are continuously monitored for behavioral and temporal conformance, while a Simulation Action Controller generates targeted interventions (e.g., priority changes, rerouting, maintenance actions) and evaluates their effects in advance within a discrete-event simulation environment. The results demonstrate that this reduces implementation risks and enables the systematic testing of rare deviations. At the same time, the authors point to current limitations, such as the rule-based selection of interventions and the restricted quantitative assessment of benefits and highlight the need for dynamic action spaces (e.g., reinforcement learning) as well as more in-depth analyses of data quality.

3. Applying Process Mining in Production Systems: An Industrial Use Case

Following a prior contribution [29], which identified the requirements and outlined the conceptual foundations for linking process mining and simulation, including an initial overarching integration framework, the present study advances the approach by applying and empirically testing it in an industrial case setting. The analysis concentrates on a predominantly linear section of a production system characterized by largely automated process steps. The objective is to demonstrate the practical feasibility of the approach under real data conditions and to provide initial evidence regarding the quality of simulation-relevant derivations.

The methodological approach is outlined in Figure 1 and will be gradually refined in future studies through application to additional, increasingly complex case examples. Process mining is deliberately applied in the early phases, from data acquisition and preparation to model formalization, precisely in those stages that the literature consistently describes as particularly resource-intensive [6,30]. At the same time, the essential core steps of data collection, preprocessing, system analysis, and model formalization remain compatible with VDI 3633 [31] as shown in Figure 2. The contribution positions itself as a data-driven specification of this established methodology [highlighted in color]. The following sections provide a detailed account of the approach and the concrete use case.

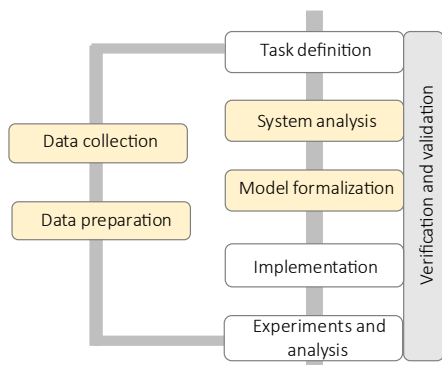


Figure 2: Procedure model for a simulation, according to VDI 3633 [31]

3.1. Data acquisition, cleansing and formalization

The data used for the case study were obtained from a production-related MES. For data protection reasons, they were pseudonymized and, in parts, structurally modified; process-specific details are therefore deliberately not reported. In this initial investigation, the data basis is deliberately restricted to MES event logs to address the question of whether material

flows can be sufficiently reconstructed at this higher system level and rendered usable for simulation.

The examined segment (see Figure 3) comprises ten stations (M1–M10) and three product groups (A, B, C) over an observation period of two weeks. Each order (case) is assigned a unique case ID and a start timestamp. Every event is associated with an event type that marks the entry (“In”) or exit (“Out”) of a process. For the analysis, the process mining software Disco, which is widely applied in academic contexts, as well as Celonis, primarily used in industrial practice, were employed. Methodologically, the processing requires that both start and end timestamps for each activity are available within the same dataset to correctly determine throughput times.

CaseID	TimestampStart	Product	Machine	Event type
28	02.09.2024 06:20	A9	M1	IN
29	02.09.2024 06:20	A3	M2	Out
30	02.09.2024 06:20	A4	M2	IN
31	02.09.2024 06:20	A3	M3	IN
32	02.09.2024 06:22	A2	M5	Out
33	02.09.2024 06:22	A3	M3	Out
34	02.09.2024 06:22	A9	M1	Out
35	02.09.2024 06:22	A10	M1	IN

Figure 3: Excerpt from the event data initially transferred

During data preprocessing, issues of data quality emerged, such as multiple recorded exits (“Out”) at different timestamps. These inconsistencies were addressed by applying a deterministic cleaning rule: for each case and process, only the earliest valid exit timestamp was retained, while redundant subsequent entries were removed as duplicates. Event cleaning was thus carried out according to temporal consistency ($\text{start} \leq \text{end}$), ensuring a monotonic sequencing of process steps within each case; conspicuous instances were flagged or excluded from further analysis. All transformation steps and cleaning rules were documented to guarantee the traceability and reproducibility of the data preparation.

3.2. Data and system analysis and model formalization

After import into the process mining software, the control flow was automatically discovered (see Figure 4) and annotated with performance information. The results were then subjected to analytical interpretation. The initial performance annotation revealed markedly inflated activity durations. Distributional analyses showed systematic distortions, particularly in event pairs recorded across shift boundaries. For example, activities initiated on Friday evening but registered as completed only on Monday were interpreted under the initial assumption of a 24/7 calendar as continuous processing time. As a result, weekends and night periods were erroneously

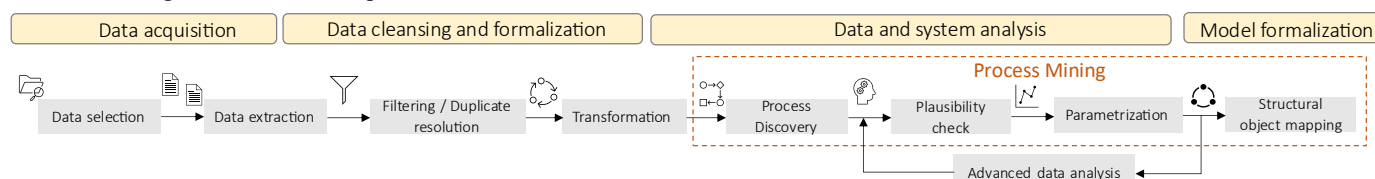


Figure 1: Procedure for using process mining for parts of simulation development

included in the cycle time calculation, leading to excessively inflated throughput times of up to 56 hours. To correct these effects, an explicit shift calendar with two weekday shifts from 06:00 to 22:00 was configured manually within the software. The corresponding shift pattern was inferred directly from the event log by analyzing timestamp distributions, which revealed recurring clusters of activity between 06:00 and 22:00 and systematic idle periods overnight and on weekends. Following reannotation based on the shift calendar, off hours were excluded from processing times, and events spanning shift or weekend boundaries were adjusted in a calendar aware manner. This resulted in substantial changes to means, percentiles and variances of process durations, bringing them into a plausible relation to the observed operations. For the subsequent parameterization of the simulation, this implies that activity times must be converted into working time normalized processing durations, while downtimes and periods of unavailability must be explicitly represented through shift calendars and resource availabilities.

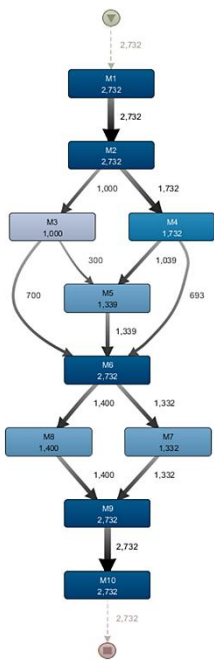


Figure 4: Process flow generated by process mining

Processing Times

To adequately represent the processing times relevant for modeling, further statistical analysis is required. By fitting different distributions to the empirical data, it is possible not only to capture the central tendency but also to account for the characteristic skewness and variance of the time distributions. An exemplary analysis is presented for machine M2. The empirical processing times recorded at machine M2 exhibit considerable dispersion that cannot be sufficiently described by a normal distribution. Estimating a normal distribution highlights the inadequacy of this assumption, as the high variance prevents it from realistically reflecting the observed data. A better fit was obtained through approximation with skewed distributions. In particular, the lognormal and gamma distributions provided a more realistic representation of the

empirical data, since processing times are by definition non-negative and often display right-skewed behavior. For the lognormal distribution,

$$X \sim \text{Lognormal}(\mu, \sigma), f(x) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln x - \mu)^2}{2\sigma^2}\right), x > 0.$$

a shape parameter of $\sigma = 0.39$ and a scale parameter of $e^\mu = 162.3$ s were estimated.

Disregarding external disturbances such as machine downtimes or setup times, the fitted distributions suggest that the typical processing time for a workpiece at machine M2 is most likely around two minutes and thirty seconds. This value corresponds to both the median and the mode of the estimated distributions and thus represents a realistic expectation for undisturbed operation. For simulation purposes, the use of a lognormal or gamma distribution is therefore recommended, as both adequately capture the empirical skewness and variance of the data. However, this choice requires validation within the simulation environment.

Transport time and buffer capacities

The observed intermediate times Δt can initially be explored using standard process mining tools. Both systems provide the functionality to visualize processing or transition times as histograms or distributions and to compute descriptive statistics such as medians, percentiles, or extreme values. This allows for an initial assessment of whether the time differences are relatively constant, highly variable, or shaped by outliers. In the present analysis, the example of the deltas measured between “M1: OUT” and “M2: IN” is used as process edges within the tools, supplemented by their frequency distribution. For the parameterization of a buffer in a simulation model, however, a more in-depth analysis is required that goes beyond mere visualization. In the proposed approach, this gap is addressed by decomposing the intermediate times Δt into a deterministic component T (transport) and a stochastic component W (waiting). While Disco and Celonis provide a first orientation regarding orders of magnitude and outliers, the decomposition $\Delta t = T + W$ establishes the basis for realistic modeling. Transport times can thus be implemented as deterministic delays, whereas waiting times emerge endogenously in the simulation through resource logic and buffer capacity.

To estimate the transport time, the state of the buffer at the moment of arrival is considered. If no backlog is present, $Q(t_{arr}) = 0$, the arriving part can be taken over immediately by M2. In these cases, Δt corresponds solely to the transport time T . Analysis of these observations shows that the transport time is constant at approximately $T \approx 2$ s. All other observations with $\Delta t > T$ comprise, in addition to the transport time, waiting times caused by backlog in the buffer. Thus, long intermediate times of up to several minutes are not attributable to physical transport but rather to waiting phases resulting from the occupation of M2.

For the simulation, the relevant transport time is therefore modeled deterministically with $T = 2$ s. Waiting times W , by contrast, do not arise exogenously but endogenously within the simulation model through the interaction of processing time variability and the available buffer capacity. To determine this capacity, the time-dependent buffer process $Q(t)$ was reconstructed: each arrival from M1 after transport increases

the stock by one, and each start at M2 decreases it by one. From the resulting trajectory, the time-weighted distribution of buffer levels was derived. The results indicate a median buffer occupancy of four parts, a 90th percentile of five parts, and 95th and 99th percentiles of six parts, with a maximum observed level of eight parts. For a dimensioning at a 95% service level, a recommended buffer size of six places can thus be derived. This ensures that backlogs are avoided in 95% of the observed time.

Setup time

In the detailed analysis of machine M2, the event data were also examined specifically regarding product changes. Common process mining tools such as Disco or Celonis allow the visualization of throughput and transition times as well as the identification of striking outliers, but they do not provide a differentiated separation of waiting times and, in this case, setup-related times. For the evaluation, two time categories were distinguished:

1. the waiting time between the completion of product A and the start of the subsequent product B ($Out_A \rightarrow In_B$)
2. the processing time of the first part of a new product group after the change (difference between In_B and OUT_B).

The results show that waiting times at the moment of change are practically negligible (median and 90th percentile at 0 s, mean 19 s). By contrast, the processing time of the first part after a product change increases significantly (mean 319 s, median and 90th percentile 150 s).

This indicates that setup times do not appear as preceding idle periods but are embedded in the extended processing time of the first workpiece following a product change. For simulation purposes, this should be represented by modeling the transport time deterministically and accounting for setup through product-dependent extensions of the processing time of the first item. In the present application, model formalization could be carried out in a comparatively straightforward manner, since all process steps under consideration were uniquely assigned to a single resource. Consequently, the control flow reconstructed from the event logs can be directly mapped onto resource modules in the simulation model without additional abstraction. More complex scenarios with competing or parallel resources, by contrast, require an extension towards an object-centric approach in which relationships between objects (e.g., products, orders, resources) are explicitly modeled to correctly represent multiple assignments or shared resource usage.

3.3. Validation

Based on the information extracted from the event log regarding process structure, task sequence, and statistically evaluated parameters, a simulation model of the production stage under consideration was developed in Siemens Plant Simulation. The validation of this model is carried out by comparing the performance indicators generated by the simulation with the reference values derived from the raw data. The central validation criteria include the distributions of processing times, with explicit consideration of the extended processing times after product changes, the time-weighted buffer levels between machines, the resulting waiting times as

emergent behavior from resource logic and buffer capacity, as well as system-level indicators such as throughput and resource utilization. The comparison of key indicators is presented in Table 1.

Table 1. Comparison of Key Performance Indicators: Event Log vs. Simulation.

KPIs	from event logs	from simulation
Output	2752	2744
Cycle time per part	3:29	3:31
Average lead time	36:47	38:45

Resource Statistics - Resource Statistics

Object	Working	Set-up	Waiting	Blocked	Powering up/down	Failed	Stopped	Paused	Unplanned	Portion
M2	48.60%	27.81%	7.06%	0.00%	0.00%	16.53%	0.00%	0.00%	0.00%	

Portions of the States

Object	Maximum Contents	Relative Empty	Relative Full	Relative Occupation without Interruptions	Relative Occupation with Interruptions
Buffer	4	13.96%	0.00%	0.00%	0.00%

Material Flow Properties

Figure 5: Excerpt from the statistics report from Plant Simulation for machine M2 and the upstream buffer

The simulation results for the states of M2 confirm the distribution parameters previously derived from the log data. With a working time share of 48.6% and a setup share of 27.8%, the model reflects that the extended processing times of the first part after a product change make a substantial contribution to the overall workload. This finding is consistent with the analysis of processing time distributions, which systematically demonstrated increased durations for the first parts of a product group. The additionally observed waiting time share of 7.1% can be attributed to the interaction between processing time variability and buffer capacity, thereby supporting the previously assumed decomposition of intermediate times into transport and waiting components.

The buffer statistics likewise confirm the capacity derived from the distributional analysis: with a maximum occupancy of four parts and an empty-state share of 14%, the 95th percentile of the buffer analysis ($\approx$ six parts) is not exceeded, indicating that the chosen capacity provides sufficient reserves. At the same time, the moderate utilization shows that the constant transport time of two seconds and the endogenously generated waiting times are represented plausibly within the simulation.

4. Conclusion and Outlook

The case study demonstrates that the combination of process mining and simulation makes it possible to construct a consistent simulation model even under limited data availability, one that realistically captures key process characteristics such as processing times, setup logic, and buffer dynamics. In particular, the analysis of intermediate times and their decomposition into deterministic transport and stochastic waiting components enabled a robust parameterization of the model. This confirms that process mining outcomes can provide not only structural but also differentiated temporal foundations for simulation.

At the same time, the application highlights methodological limitations. Process mining already requires targeted preprocessing of raw data, as these must be transformed into an

event log format suitable for analysis. This involves initial steps of data cleaning and mapping, conducted outside standard process mining tools, to ensure the unambiguous referencing of cases and activities. Within the analysis itself, further constraints become apparent: tools such as Disco or Celonis support the exploration of control flows and the visualization of empirical distributions for parameterization, but they do not provide functionalities for distribution fitting or for decomposing times into transport, setup, and waiting components as required in production processes. This methodological gap necessitates additional statistical analyses that must be conducted outside the standard tool environment.

From these findings, several directions for future research emerge. The approach should be extended to more complex scenarios with multiple resources and intertwined material flows; although the present case focuses on single-piece flow, it is in principle extendable to batch processes, hierarchical products, and operator-dependent constraints, provided that event data contain suitable identifiers (e.g., batch IDs, component–assembly relations, operator assignments), an extension that can be supported by object-centric process mining. Moreover, process mining tools need to be enhanced and more tightly integrated with simulation environments, for example through standardized interfaces. Finally, validation should not only rely on performance indicators such as throughput, utilization, and buffer levels, but also consider event-level process behavior to ensure consistency between simulated and real processes. In this way, the study contributes to the data-driven design of production systems by systematically leveraging event logs for model construction and validation.

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